University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,864: Ubiquitous Computing

ECTS credits: 3

Overview examination/s

(binding regulations see below)
decentral - Oral examination and technical discussions, Analog, Individual work individual grade (70%)
Examination time: Term time
decentral - Programming, Digital, Individual work individual grade (30%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,864,1.00 Ubiquitous Computing -- English -- Mayer Simon
7,864,2.00 Ubiquitous Computing: Exercises -- English -- Bekavac Luka , Strecker-Bischoff Jannis

Course information

Prerequisites

- For computer science students: Interest in the approaches behind and implications of the permeation of people's lives with computing technology
- For students from outside of core computer science: Motivation to engage with technology and to understand the integration of technical, business, and societal factors

Learning objectives

Overarching Learning Goal: Students know the history, cross-disciplinary drivers, and current state of the field of Ubiquitous Computing. They recognize patterns, can evaluate technology-driven business models, and can make educated estimates regarding future trends in the post-PC age.

Overarching Competency Goal: Students are capable of thinking of technology as embedded in individual and societal structures. They are able to zoom out of an individual, narrow perspective on technology solutions and to rather consider technology in its ecological embedding.

Learning Goals (Knowledge)

- Students know what *resource-constrainedness* (e.g., battery, communication channels, computing power) means in the context of Ubiquitous Computing and can recognize resource-constrained contexts when they encounter them. They are aware of methods and approaches that are suitable in resource-constrained contexts, in particular regarding sensing and identification.
- Students know about current indoor and outdoor *localization* systems and the underlying localization principles (triangulation, trilateration, multilateration, fingerprinting).
- Students know about the current scope and limitations of *Mixed Reality* technologies and of (mobile) gaze tracking, and are aware of the expected developments in these domains.
- Students understand how these aspects interact in Ubiquitous Computing systems with respect to handling of the resulting data streams.
- Students have a good grasp of the *societal and legal context* of the development towards Ubiquitous Computing, and can evaluate Ubiquitous Computing systems in this regard.

Learning Goals (Competency)

- Students are able to implement small functional Ubiquitous Computing prototypes.
- Students are able to take informed decisions on Ubiquitous Computing projects.
- Students can act upon the technology implications of business decisions in this domain, and vice versa.

Content

This course covers fundamental **concepts, technologies, drivers, trends, and implications of Ubiquitous Computing**. The course gives both, an overview of UbiComp methods and deeper dives into foundational technologies of UbiComp. In addition, we discuss implications of the proliferation of UbiComp on individuals, businesses, and on society as a whole.

This course is designed as an **integration vessel between computer science and non-computer science students around the topic of Ubiquitous Computing** and how it creates technology, business, legal, and societal opportunities and challenges. The course stretches over the **first six weeks of the semester**, while the individual exams take place during the second half of the semester - we will coordinate the exam dates bilaterally. To allow computer science students to dive deeper into the topic, we highly recommend attending the course "**Ubiquitous Computing: Seminar and Project**", which will take place in the second half of the semester, in addition to this course.

Structure and indications of the learning and teaching design

The Ubiquitous Computing course features **lectures** that are integrated with multi-week **hands-on exercise projects**. The course elements are designed to convey theory as well as hands-on experience.

The course stretches over the **first six weeks of the semester**. After an introduction to UbiComp and its drivers, the course focuses on **four central aspects** of modern UbiComp systems: Low-Power Identification and Sensing, Context and Location, Human-in-the-Loop Methods, and Societal Integration. Students gain experience using some of the introduced technologies in the context of **three exercises** on Low-Power Sensing, Gaze-enabled Mixed Reality, and Decentralized Data Storage and Processing.

Literature

The course is based on **current research and practice results**, and relevant publications and supplementary material from the years 2020-2025 are referenced from the course materials. We encourage students to engage as well with contextual literature, including:

- John Krumm (Ed.): Ubiquitous Computing Fundamentals. Taylor & Francis, 2009
- Friedemann Mattern (Ed.): Die Informatisierung des Alltags. Springer-Verlag, 2007

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Oral examination and technical discussions
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	70%
Duration	--

Languages

Question language: English
Answer language: English

Remark
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Examination-aid rule
Closed Book

The use of aids is prohibited as a matter of principle, with the exception of pocket calculator models of the Texas Instruments TI-30 series and, in case of non-language exams, bilingual dictionaries without any handwritten notes. Any other aids that are admissible must be explicitly listed by faculty members in the paragraph entitled “Supplementary aids” of the course and examination fact sheet; this list is exhaustive.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

No further aids are permitted.

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2. Examination sub part (2/2)

Examination modalities

Examination type	Programming
Responsible for organisation	decentral
Examination form	Practical test
Examination mode	Digital
Time of examination	Term time
Examination execution	Synchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	30%
Duration	--

Languages
Question language: English
Answer language: English

Remark
Competency-adapted implementation exercises

Examination-aid rule
Open Book

Students are free to choose aids, apart from the following restrictions:

- pocket calculator models which are not part of the Texas Instruments TI-30 series, as well as any programmable electronic devices that are capable of communication such as electronic dictionaries, notebooks, tablets, smartphones, headsets, additional screens, etc. are not admissible;
- there is an option for faculty members to explicitly define exceptions under supplementary aids.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

Students are required to appropriately reference any special aids they use while working on their projects and assignments.

Content

Oral Exam (70%): The oral exam covers the content of the lectures as well as in-class discussions and referenced materials.

Exercises (30%): The lecture is accompanied by three exercises of two weeks each. Each exercise is graded binary (pass/fail).

Literature

Lecture and exercise slides, exercise assignments, provided hand-outs, and referenced literature during the course.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).